UML COMPONENT DIAGRAM — DIGITAL DOCUMENT VERIFICATION SYSTEM

Software Engineering Practical Lab 08 | Logical Components, Provided/Required Interfaces & Dependencies

Presentation Tier

«component»
User Interface / Web Client
- Verifier Public Portal (ID, QR & File Upload)
- Issuing Organization Issuance & Management Console
- Administrator Governance & Audit Dashboard

«use» IAccountService
«use» IRegistrationService
«use» IVerificationService
«use» ILogService

Application / Business Logic Tier

«component»
Account Management
- User Registration & Authentication (FR-01)
- Password Hashing & Policy Enforcer (FR-04)
- Organization Approval & Role Control (FR-07)

«component»
Document Registration
- Metadata Ingestion & File Type Filter (FR-20)
- Verification ID & QR Code Generator (FR-15)
- Document Issuance Orchestration (FR-13)

«component»
Document Verification
- Multi-channel Query Dispatcher (FR-22–FR-2
- Integrity Comparison Coordinator (FR-33)
- Status Evaluator & Verdict Producer (FR-27)

«use» ILogService

«component»
Verification Log Management
- Guaranteed Verification Audit Logger (FR-37)
- Personal History Ledger Provider (FR-39)
- System-wide Filter & Compliance Reporter (FR-40)

«use» IHashService
«use» IHashService

«component»
Integrity Hash Processing
- Standard Cryptographic Engine (SHA-256)
- am Digest Generator (FR-1
- Zero-State Independent Hash Verification (FR-34)

«use» IDataAccess
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Data Persistence Tier

«component»
Data Persistence / Database Access
- UserStore Data Access Object (DAO) (D1 Interface)
- DocumentStore Data Access Object (DAO) (D2 Interface)
- VerificationLogStore Data Access Object (DAO) (D3 Interface)
- Connection Pooling, Transactions & Referential Integrity Enforcement